

Math Nuggets for GEN AI

Binary Cross-Entropy (BCE) Loss and its Connection to Maximum Likelihood Estimation (MLE)

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September 19, 2025

Introduction

Binary Cross-Entropy (BCE) is a fundamental loss function used in binary classification tasks, particularly in deep learning and logistic regression. It measures the difference between the predicted probabilities and the actual class labels.

1. Binary Cross-Entropy Loss

Given:

- True label $y \in \{0, 1\}$
- Model prediction $\hat{y} = P(y = 1|x; \theta)$ (output of a sigmoid function)

The BCE loss for a single data point is:

$$\mathcal{L}(y, \hat{y}) = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (1)$$

For N samples:

$$\mathcal{L}_{total} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (2)$$

Intuition

- If $y = 1$, loss is $-\log(\hat{y})$, penalizing low predicted probability for the positive class.
- If $y = 0$, loss is $-\log(1 - \hat{y})$, penalizing high predicted probability for the negative class.

2. Connection to Maximum Likelihood Estimation (MLE)

BCE arises naturally when assuming that data is generated from a Bernoulli distribution.

Step 1: Bernoulli Likelihood

$$P(y|x; \theta) = \hat{y}^y (1 - \hat{y})^{(1-y)} \quad (3)$$

Step 2: Likelihood for N samples

$$\mathcal{L}(\theta) = \prod_{i=1}^N \hat{y}_i^{y_i} (1 - \hat{y}_i)^{(1-y_i)} \quad (4)$$

Step 3: Log-Likelihood

$$\log \mathcal{L}(\theta) = \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (5)$$

Step 4: Negative Log-Likelihood (NLL)

To maximize likelihood, minimize the negative log-likelihood:

$$-\log \mathcal{L}(\theta) = - \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (6)$$

This is exactly the BCE loss (apart from normalization by N).

Key Insight

Minimizing BCE is mathematically equivalent to performing MLE for a Bernoulli-distributed target variable. Thus, BCE serves as a principled and theoretically grounded loss function.